Preserving Digital Art – case study on Video Flag by Nam June Paik

The preservation of cultural heritage has long been a central concern for museums, archivists, and conservation professionals. Through decades of research and practice, conservation has established a robust set of principles and techniques for the care of traditional artworks and artifacts. However, the emerge of digital technologies in artistic creation has fundamentally changed the field of conservation. Traditional artworks, such as paintings and sculptures, generally possess a degree of stability that allows them to endure for long periods when stored and displayed under appropriate conditions. Their deterioration processes are relatively well understood, enabling conservators to monitor their condition, mitigate risks, and intervene when necessary to ensure their long-term survival while staying faithful to the creator's original intentions.

In contrast, digital artworks are far more vulnerable thanks to technological obsolescence and material degradation, since the art is inseparable from its own technological system. Software, hardware, file formats, and storage media can become outdated within a few years, making access to digital pieces increasingly difficult. As a result, a digital artwork may require conservation interventions within less than a decade of its creation in order to remain accessible and retain its original form, posing new and complex challenges for the preservation of cultural heritage. A work created today may become inaccessible within a few years if the technology required becomes obsolete, needing a lot of effort to be properly conserved. This reality challenges many of the assumptions that have traditionally guided conservation practice. For these reasons the conservation of digital art requires approaches that differ significantly from those developed for physical collections[1].

Should we prioritize the original hardware or the viewer experience? Usually we tend to follow the second option, but every case is different and needs his own studies.

We now have preventive measures to store, move and display this new type of art. Conservators try to minimize interventions, and, if necessary, they ensure that they are reversible and in line with the original form and idea. However, the preservation of born-digital artworks presents a unique challenge: some degree of change is often unavoidable. Whether through migration, emulation, or the replacement of obsolete technologies, conservation actions inevitably alter aspects of the work. This raises important ethical questions regarding authenticity and artistic intent, while also having potential aesthetic consequences that can influence how audiences experience, interpret, and respond to the artwork[2]. Today the preservation of the artist-intended behaviors may be privileged over the general technical constituents, meaning that some original components will be replaced with never technologies, while maintain the aesthetic of the piece. The choices of technologies and

[1] Real, William A. "Toward Guidelines for Practice in the Preservation and Documentation of Technology-Based Installation Art." Journal of the American Institute for Conservation 40, no. 3 (2001): 211–231.
[2] Balch, Patricia. "Electronic Media: Rethinking the Conservator's Role—Proceedings." In Modern Art: Who Cares? An Interdisciplinary Research Project and an International Symposium on the Conservation of Modern and Contemporary Art, edited by IJsbrand Hummelen and Dionne Sillé, 300–307.

programming languages in these types of art can be equated to a painter choices in color medium, since analyzing the original code helps experts discovers unique insights [3].

This type of conservation aims to identify and monitor the degree of change possible without damaging the initial idea, so the conservator works closely with experts like media technicians and video engineers. In the last decade a lot of studies were conducted on this argument, but there is not a lot of actual projects. Most of the professional in this field work collaboratively across institutions to expand their knowledge and expertise, creating a global network of specialist.

The scheduled exhibition of a media artwork is especially important for conservator action, since the art object only actually exist when installed. This gives the opportunity to conservators to gain knowledge about the conservation of the piece and its vulnerabilities[4].

Over the last few decades video installations and projections are becoming more and more common, showing us that the problem of conservation not only is real, but it's urgent. In some cases we can still find videotapes simply stored, without somebody who periodically check on them to see if they still work or need an imminent hardware update[5].

In this paper we are going to analyze *Video Flag (1996),* as a case study in the conservation of born-digital and time-based media art. Made by Nam June Paik, an artist commonly seen as the father of the video art. Throughout his career, Paik challenged conventional understandings of artistic media by incorporating emerging technologies into his practice[6]. He tried to humanize technology and electronic media, via his writing and his art, demystifying the medium[7]. Through manipulated television sets, live performances, video installations, and satellite broadcasts, he explored the cultural and social implications of electronic communication, introducing the use of video as a form of artistic expression [8]. He was also an important member of the Fluxus movement, an international avant-garde network founded in the '60s, that wanted to revolutionize art and transform quotidian actions into something more, influencing what we now define as art [9].

A recurring theme in Paik's work is the relationship between technology, culture, and the circulation of information. He is known for fusing new technologies, creating suggestive and simultaneously unusual work about the construction of culture, usually focusing on the movement of information and the television, a tool that he thought was a new way to create

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[3] Engel, Deena, and Joanna Phillips. "Applying Conservation Ethics to the Examination and Treatment of Software- and Computer-Based Art." Journal of the American Institute for Conservation 58, no. 3 (2019): 180–195.
[4] Guggenheim Museum. "Time-Based Media Conservation."
[5] Balch, Patricia. "Electronic Media: Rethinking the Conservator's Role—Proceedings." In Modern Art: Who Cares? An Interdisciplinary Research Project and an International Symposium on the Conservation of Modern and Contemporary Art, edited by IJsbrand Hummelen and Dionne Sillé, 300–307.
[6] Guggenheim Museum. "Nam June Paik, TV Buddha."
[7] Weintraub, Linda. "Traversing the Worlds of Nam June Paik." Sculpture Magazine.
[8] Guggenheim Museum. "Nam June Paik, TV Buddha."
[9] Tate. "Fluxus."

education and keep memory alive[10]. He viewed television not merely as a medium of entertainment but as a powerful tool capable of transforming education, communication, and collective memory.

In "*Expanded Education for the Paperless Society*", that he published in 1970, he wrote: “Therefore, nothing is more urgent and successproof than to film the images and voices of aging great thinkers of today, and yesterday, in sufficient and surplus quantity, who might pass away any day, such as Marcel Duchamp, Jaspers, Heidegger…”. Not only in his paper he mentions great thinkers, but also philosophy, reading aids, instant global university (in a way creating a percussor of today digital universities), singing lessons, archive, audio tape library, imaging a future where the global access of information was a reality[11]. In a pre-internet era, he predicted the importance of technology in the global network of immediate communication, anticipating many technological developments that are now almost taken for granted[12].

In his early works he began experimenting with altered television, manipulating signals with magnets and using the video feedback, synthesizers and other technology to create abstract shapes, luminous colors and changing visuals. Over time, these investigations evolved into increasingly complex sculptural arrangements of television monitors, in which electronic images became both the medium and the material of the artwork.

Video Flag (1996) is part of a longer series of other artwork with the American flag (“*Video Flag X”, “Video Flag Y”* and *“Video Flag Z”* all made in the middle of the ‘80s, composed of 84 10” CRT monitors in a 7 x 12 grid). Unlike the earlier versions produced during the 1980s, the 1996 installation reflects significant technological developments that occurred during the following decade. It incorporates LaserDisc players and custom switchers that can be programmed to merge the four video channels into a collage of split-second images and binary numbers. The artwork is a monolithic unit that represents the American flag’s stars and stripes, but instead of using the flag a fixed national symbol, it uses electronic programming and fast paced edits with psychedelic effects of iconic moments from American politics, like a morphing sequence representing U.S. president portraits, from Truman to Clinton, pieces of the star, news stills… The flag functions as the backdrop against the individual cultural events, personas and technologies are presented to the viewer. The work invites viewers to reflect on the relationship between national identity and mass communication, while simultaneously demonstrating how television and digital technologies shape the way history and culture are produced, transmitted, and remembered.

Like many other born-digital artwork *Video Flag* presents significant conservation challenges. As we said before, preserving these types of works is a lot more complicated than it seems,

[10] Tate. “Nam June Paik.”
[11] Nam June Paik, "Expanded Education for the Paperless Society," Radical Software 1, no. 1 (Summer 1970): 9
[12] Tate. “Nam June Paik.”

especially when comparing them with traditional objects. Technologies improves at a very rapid speed, and the equipment cannot stand the test of time.

In this specific case the Hirshhorn Museum and Sculpture Garden has put in place several interventions to preserve the long-term functionality of the piece. At first they migrated the files from the original LaserDisc players to DVD players, with the help of an electronics specialist. Changing the disc storage medium was a fundamental step to ensure longevity, and in the near future there is a good probability that the medium will be replaced again, following the fast evolution of today's technologies. Over time they implemented the use of fans, rewiring and television assessment to increase the durability of the piece. But the real conservation work took place between 2015-2019. In case of complex piece like this one, every element and behavior needs to be studied, to understand his work inside the complexity of the installation. Given the complexity of the piece, experts decided to detect three main phases, as we can read on the Smithsonian website: "physical assessment and preparation for display; addressing media technologies and their potential obsolescence and migration; and storage and preparation for loan or travel".

The basic armature of the artwork allows access to the backs of the structure, for installation, maintenance and wiring. The CRT – cathode ray tube – were heavily modified by June Paik to give space for a metal housing, creating the possibility to stack the monitors in a 7 x 10 grid, resulting in a massive final wall of screens, 70 in total. The visual composition is generated through four independent video channels, which are currently played from DVD sources and routed through a series of custom electronic devices. These devices control the rapid switching and distribution of images across the monitors, producing the characteristic flashing patterns that define the work. The arrangement of the screens replicates the structure of the American flag. Sixteen monitors positioned in the upper-right section display star imagery through a modified commercial multi-screen device capable of distributing the video signal in multiple configurations. The red and white stripes are formed by rapidly alternating sequences of imagery, including streams of binary numbers and other video content, controlled through custom switchers. The resulting visual effect is one of constant movement and transformation, emphasizing the dynamic relationship between national symbolism, electronic media, and information technology.

Another important step in the conservation of the piece was taken around 2007, when it was identified as a potential fire hazard due to the closely packed monitors and the amount of generated heat. The Hirshhorn Museum collaborated with engineers, conservators and audiovisual specialists to install a new electrical infrastructure, including programmable power distribution units, environmental monitoring systems and improved ventilation. To reduce fire risks, a VESDA very-early-warning smoke detection system was integrated, automatically shutting down all equipment if smoke is detected. These upgrades also enabled remote monitoring of temperature, humidity, power consumption, and system performance throughout exhibitions. Importantly, all modifications were designed to be

reversible and removable, ensuring that the conservation intervention respected the integrity of the original artwork.

A significant conservation challenge for Video Flag concerned the heat generated by its numerous CRT monitors. Extended exposure to high temperatures can severely affect the longevity of these devices, particularly due to the electrolytic capacitors contained within their circuitry. These components rely on a liquid electrolyte that gradually evaporates over time, a process accelerated by excessive heat. As the electrolyte dries out, the capacitors lose efficiency and may eventually fail, compromising the functionality of the monitors. Because the CRTs used in the installation are both heavily modified and increasingly difficult to replace, preventing heat-related deterioration became a priority for conservators.

The artwork's armature was designed with eight openings in the upper rear enclosure to accommodate ventilation fans. In the early 2000s, technician Dan Meijer replaced the original fans with higher-quality models, an intervention that significantly reduced monitor failures. During a later assessment in 2017, conservators concluded that further improvements in airflow could enhance the long-term stability of the installation. Any modifications, however, needed to respect the visual appearance and operational behavior of the artwork. Additional cooling measures were therefore implemented without altering the structure of the armature, balancing the preservation of the technological components with the conservation principle of maintaining the integrity of the artist's original work[13].

The conservation of Video Flag demonstrates how preserving contemporary media artworks often requires a balance between authenticity and technological adaptation. Every intervention, from media migration to hardware replacement, raises questions about how much change can be introduced before the identity of the artwork itself is altered. For conservators, the challenge is not simply to preserve the physical components of the time-based media, but to maintain the visual behavior, functionality, and conceptual meaning that define the work as originally intended by the artist, choosing what to preserve and what can be modified to ensure the survival of the piece.

As technology continues to evolve, the conservation of works such as *Video Flag* demonstrates the need for flexible preservation strategies that balance technological adaptation with the preservation of artistic intent and authenticity.

[13] Smithsonian American Art Museum. “Video Flag.” Time-Based Media Art Initiative.

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